Verify Numbers and Dates

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

Verify Numbers and Dates

> Prepare Verify Numbers and Dates: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Explain the **mechanism** and terminology of claim support and citation evaluation

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min

> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Case exposure check: this source example appears in ai-2/29, ai-3/16. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.

Retrieval answer guidance

Source Quality

Without opening the old lesson, explain one key idea from Source Quality. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of conflicting and time-sensitive sources

Reference example, not the only acceptable wording: Teacher explanation: Changing facts require a stated time and source-version policy. Current-looking records can conflict without enough **evidence** to choose between them.

For this toy task a signed amendment supersedes the referenced original from its effective time. Keep both records. A newer timestamp alone has no authority unless the policy grants it.

Multimodal Evidence

Without opening the old lesson, explain one key idea from Multimodal **Evidence**. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**

Reference example, not the only acceptable wording: Teacher explanation: A multimodal pipeline may transcribe speech, inspect an image and generate a response.

Errors at one stage can propagate even when later stages run correctly. Track source modality, segment/time references, transcription, interpretation and final claims separately.

Compare conflicting **evidence** without assuming one modality always wins. Ask for clarification when the task needs unresolved information.

Image Generation Models

Without opening the old lesson, explain one key idea from Image Generation Models. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of image generation and provenance

Reference example, not the only acceptable wording: Teacher explanation: An image generator constructs an **output** from learned patterns and a specification. A generated picture is not a record that the depicted event occurred.

Keep a provenance record containing specification, tool/version when known, generation date, edits, permissions and intended use. Label fictional imagery.

Inspect visual requirements and avoid inferring identity or event authenticity from appearance.

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A report says 21/28=**70%** and gives February 30 as the deadline. Correct what can be computed and identify what must be checked externally.

Concept to explain

Precise values require direct **evidence** and recalculation.

Term	Meaning
primary record	An original record directly relevant to a claim, rather than a later summary of it.
recalculate	Compute a quantity again from the original values or corrected inputs.
unit	An agreed amount used to express a measurement, such as centimetres or seconds.

Term	Meaning
as-of date	The date up to which a claim or record is stated to be current.

Prepare the class

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

Alternative compact allocation

> This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Two 45-minute sessions. Citation evaluation checks source existence, exact passage support and coverage of answer claims. A real citation can still be irrelevant.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Readiness answer

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

End goal for the pictorial case

By the end, I can verify numbers and dates without inventing the intended correction.

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain the mechanism

> Model Verify Numbers and Dates using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Claim: 18 of 24 tests passed, reported as **80%**.

2. **Recalculate**: 18 ÷ 24 = 0.75 = 75%.

3. Check denominator, **unit**, rounding, and source date.

4. Correct the claim and preserve the calculation trail.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

The teaching case

An answer says 15/20=**80%** and sets a deadline of February 30.

The teaching case

Verify both.

Exact explanation to model

Why the result follows

15/20=75%. February 30 is invalid in the Gregorian calendar. Check the intended deadline from its authority rather than guessing.

Explain the added worked example

Citation evaluation checks source existence, exact passage support and coverage of answer claims. A real citation can still be irrelevant.

Explain the added worked example · 1

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

For this classroom rubric count a claim as supported only when its cited passage entails it. Separately count correct citation links. Use the number of factual claims as the support denominator.

Explain the added worked example · 2

Before applying the procedure, write its **unit**s, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Key terms

Term	Meaning
Entailment	Whether the cited passage supports the specific claim
Citation coverage	Which claims include source references
Link validity	Whether a source reference resolves to the intended record

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

An answer has four factual claims. Three have citations, but only two are supported by their cited passages. All links exist.

Claim support is 2/4=50%; citation presence is 3/4=75%; link existence among three citations is 3/3. These measure different failures.

Model the reasoning aloud · 3

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Verify Numbers and Dates

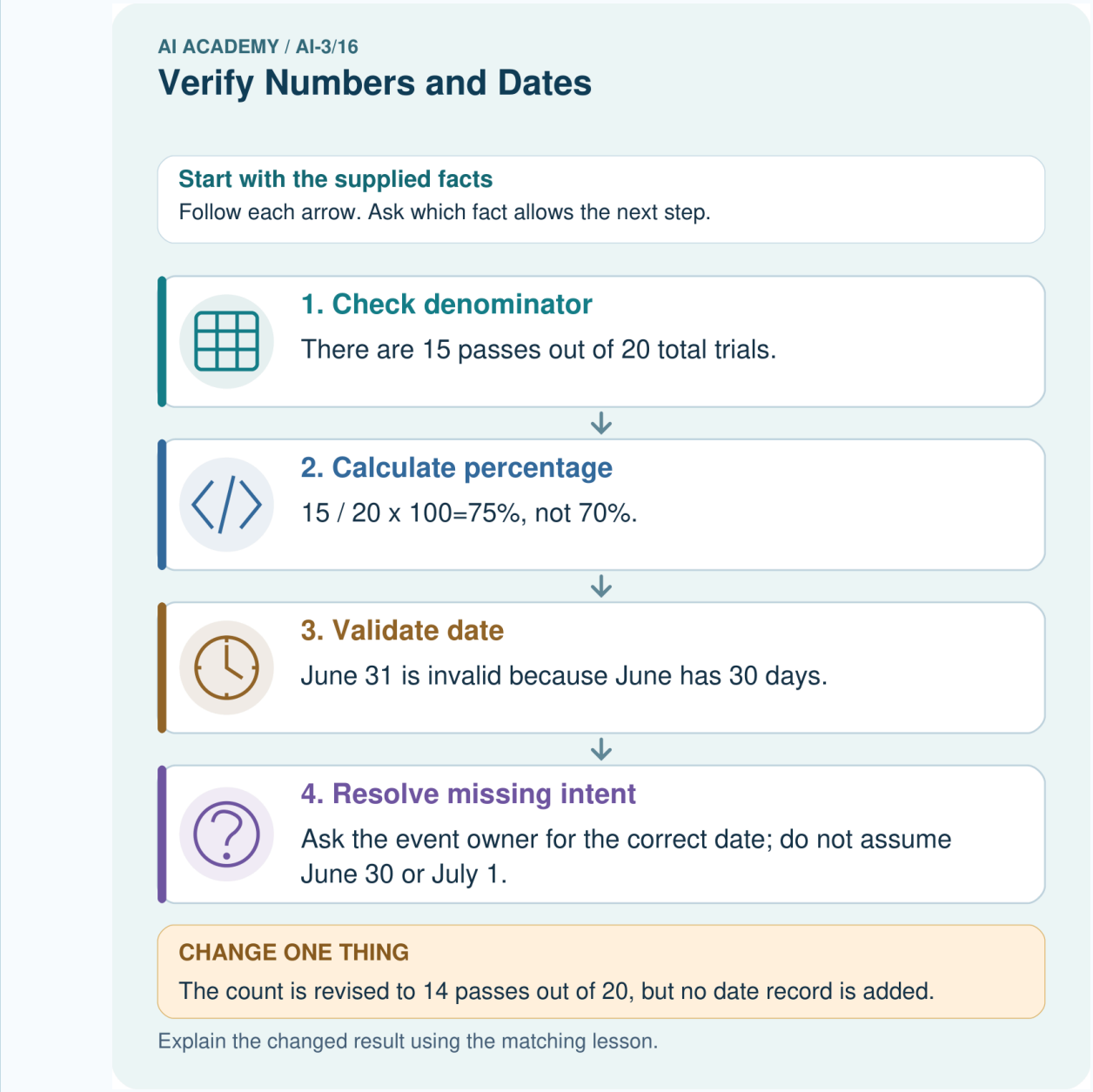
A fictional Study Buddy checks a science-club summary against supplied counts and calendar facts.

The facts we will use

> **Supplied facts**
>
> Record: 15 of 20 fictional model trials passed. Draft: "**70%** passed; the meeting is on June 31."

Supplied facts

June has 30 days. No authoritative record gives the intended meeting date.



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Check denominator	There are 15 passes out of 20 total trials.
2. Calculate percentage	15÷20×100=75%, not 70%.
3. Validate date	June 31 is invalid because June has 30 days.
4. Resolve missing intent	Ask the event owner for the correct date; do not assume June 30 or July 1.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A percentage requires the correct numerator and denominator. Calendar validation identifies an impossible date, but it cannot discover what date the author intended.

These checks use different **evidence**: arithmetic repairs the percentage from the supplied counts, while the date needs another authoritative source before publication.

What this example does not establish

Keep the stated limit

This check confirms internal arithmetic and date validity, not whether the fictional trial measurements themselves were reliable.

Change one thing in the pictured case

Change one thing

Change: The count is revised to 14 passes out of 20, but no date record is added.

Predicted result

Prediction: **70%** now matches the count; June 31 remains invalid and unresolved.

Why the result follows

Reason: 14÷20×100=**70%**; changing a count provides no new calendar **evidence**.

Ask while showing the picture

Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Verify Numbers and Dates, reduce support deliberately and check individual reasoning before discussing answers.

Board sequence

1. Put the facts on the board

Write the given facts: An answer says 15/20=**80%** and sets a deadline of February 30.

2. State the decision to explain

Write the question to explain: Verify both.

3. Reveal the worked reasoning

Reveal after **prediction**s: 15/20=75%. February 30 is invalid in the Gregorian calendar. Check the intended deadline from its authority rather than guessing.

Hold back the independent task

Keep for an independent attempt: A report says 12 of 40 responses is 40%. **Recalculate** and state the correction with the denominator.

Classroom dialogue

Ask the class

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Language to model

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write **unit**s and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

Model these guided questions

Guided: Claim support is 2/4=50%; citation presence is 3/4=75%; link existence among three citations is 3/3. These measure different failures.

Qualification: Citation counts do not establish truth beyond the supplied sources or the correctness of uncited claims.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Case-specific teaching sequence

Stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.

Stage	Teacher action
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Completion practice

Completion task: An answer has four factual claims. Three have citations, but only two are supported by their cited passages. All links exist. Show the intermediate steps and give the conclusion with its scope.

Reasoning to listen for

Teacher check: Teacher explanation: Claim support is 2/4=50%; citation presence is 3/4=75%; link existence among three citations is 3/3. These measure different failures.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Claim support and citation evaluation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Citation counts do not establish truth beyond the supplied sources or the correctness of uncited claims.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Which part can be corrected immediately from existing **evidence**, and which needs follow-up?

Reasoned teacher answer

Correct percentage to 75% in the original case. The intended date needs confirmation from the responsible event source.

Changed-case reasoning: **70%** now matches the count; June 31 remains invalid and unresolved. 14÷20×100=**70%**; changing a count provides no new calendar **evidence**.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the weekly evidence lab

> Run the Verify Numbers and Dates investigation and use the first unsupported step to choose

> a teaching response.

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

Audit six claims for source, date, **unit**, denominator, and arithmetic; mark verified, corrected, or unresolved.

> What did the hardest test teach you that the normal test did not?

Make the lab runnable

Use the concrete case as the shared normal case and the independent question as a second case. Each pair creates a boundary or missing-information case and predicts its result before testing.

Keep the expected result separate from an observed mistake.

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The worked result establishes every future case: Citation counts do not establish truth beyond the supplied sources or the correctness of uncited claims.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Verify Numbers and Dates questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Worked case answer

Teacher key · Worked case answer

15/20=75%. February 30 is invalid in the Gregorian calendar. Check the intended deadline from its authority rather than guessing.

Independent question

Independent question · withhold the key

A report says 12 of 40 responses is 40%. **Recalculate** and state the correction with the denominator.

Expected independent answer

Teacher key · Expected independent answer

12 divided by 40 equals 0.30, or 30%. The correct statement is 12 of 40 responses, 30%.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support

Criterion	0 points	1 point	2 points
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A precise-looking percentage should be accepted without calculation.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Teacher key · Original transfer answer

21/28=3/4=75%. February 30 is not a valid Gregorian date, but the intended deadline must be checked in the responsible source. Do not guess February 28 or March 2 merely because the supplied date is invalid.

Explanations for the additional workbook examples

W1 Explain the mechanism

Workbook question · W1

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher reasoning · W1

Teacher explanation: Citation evaluation checks source existence, exact passage support and coverage of answer claims. A real citation can still be irrelevant.

For this classroom rubric count a claim as supported only when its cited passage entails it. Separately count correct citation links. Use the number of factual claims as the support denominator.

W2 Trace the worked case

Workbook question · W2

An answer has four factual claims. Three have citations, but only two are supported by their cited passages. All links exist. Show the intermediate steps and give the conclusion with its scope.

Teacher reasoning · W2

Teacher explanation: Claim support is 2/4=50%; citation presence is 3/4=75%; link existence among three citations is 3/3. These measure different failures.

W3 Audit the limit

Workbook question · W3

A report omits this qualification: Citation counts do not establish truth beyond the supplied sources or the correctness of uncited claims.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher reasoning · W3

Teacher explanation: Citation counts do not establish truth beyond the supplied sources or the correctness of uncited claims.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W4 Apply a new case independently

Workbook question · W4

Five claims; four cited; three supported by cited passages; one cited ID is nonexistent. Compute support and citation presence and explain link limits.

Teacher reasoning · W4

Teacher explanation: Support=3/5=**60%**; citation presence=4/5=**80%**. Link checks reject the nonexistent ID; remaining real links still need **entailment** checks.

W5 Design a revealing follow-up

Workbook question · W5

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher reasoning · W5

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. Citation counts do not establish truth beyond the supplied sources or the correctness of uncited claims.

The new case should test this particular **mechanism** rather than merely restate the original result.

Assess independent transfer

Independent question · withhold the key

Selected case: A draft claims18 of24 learners is**60%**, and gives an April31 deadline. Check both statements without inventing a deadline.

Selected case answer

Teacher key · Selected case answer

18/24=75%. April has30 days, so April31 is invalid. Verify the intended date with an authoritative source rather than substituting April30 or May1.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect and extend learning

> Connect Verify Numbers and Dates to Verified Study Buddy, adapt support and plan a later changed-case check.

Verified Study Buddy

Route arithmetic to a calculator and dates to direct sources before presenting them.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Support and stretch

Provide one short approved passage; separate instruction, **input**, **output** and **evidence**. Stretch: introduce an unsupported claim or conflicting source and validate claim by claim with a fallback.

Exit reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Bridge to the next week

Next: Week 17: Model Memory vs Retrieval. Retrieve today's **evidence** and identify what carries forward and what is new.

Support and fresh reassessment

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

W4: Support=3/5=**60%**; citation presence=4/5=**80%**. Link checks reject the nonexistent ID; remaining real links still need **entailment** checks. Require intermediate reasoning and the stated scope.

Fresh retry: Six claims, five cited, four supported; all five links real. Expected reasoning: Support 4/6=2/3; citation presence 5/6; link existence 5/5 does not verify every claim.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Next connection: Abstention and clarification. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Plan the delayed return

Suggested return points in this level: ai-3/17, ai-3/19, ai-3/22. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

Make Study Buddy check quantities, **unit**s, dates and source support separately; record unresolved fields instead of guessing.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Verify Numbers and Dates lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Explain the **mechanism** and terminology of claim support and citation evaluation

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Sections 9 and 11–14. Give extra time to finish the full task set; use Section 15 when the week includes a coding lab.

Before class, solve the supported practice: An answer has four factual claims. Three have citations, but only two are supported by their cited passages. All links exist.

Show the intermediate steps and give the conclusion with its scope.

Supported practice answer

Reasoned practice answer: Teacher explanation: Claim support is 2/4=50%; citation presence is 3/4=75%; link existence among three citations is 3/3. These measure different failures.

Check every learner before discussion: Use the worked case for Claim support and citation evaluation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Citation counts do not establish truth beyond the supplied sources or the correctness of uncited claims.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: The count is revised to 14 passes out of 20, but no date record is added. Require a **prediction** and a reason before comparing. Expected result: **70%** now matches the count; June 31 remains invalid and unresolved.

Reason: 14÷20×100=**70%**; changing a count provides no new calendar **evidence**.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words

Criterion	Not yet	With support	Independent
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

For later coding lessons

> This week has no code block. Use this routine when a later lesson includes an executable lab.

> Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

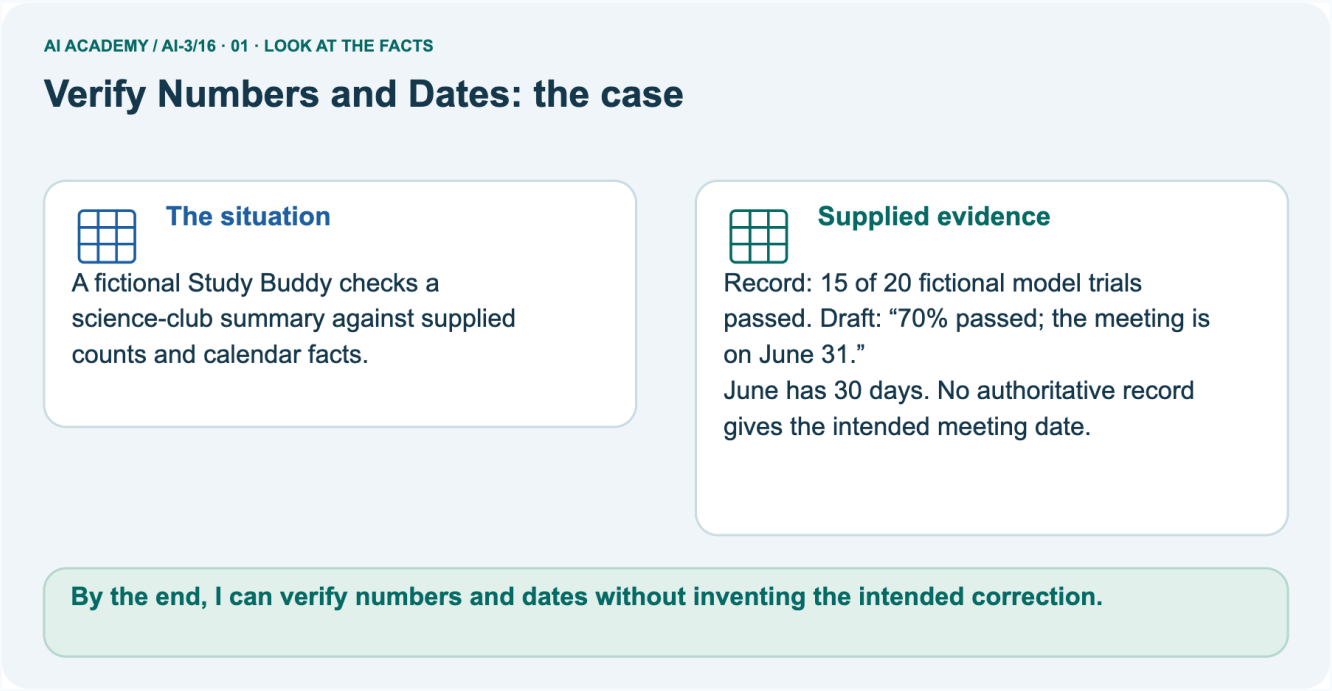
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Verify Numbers and Dates graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Verify Numbers and Dates: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional Study Buddy checks a science-club summary against supplied counts and calendar facts. Supplied facts: Record: 15 of 20 fictional model trials passed.

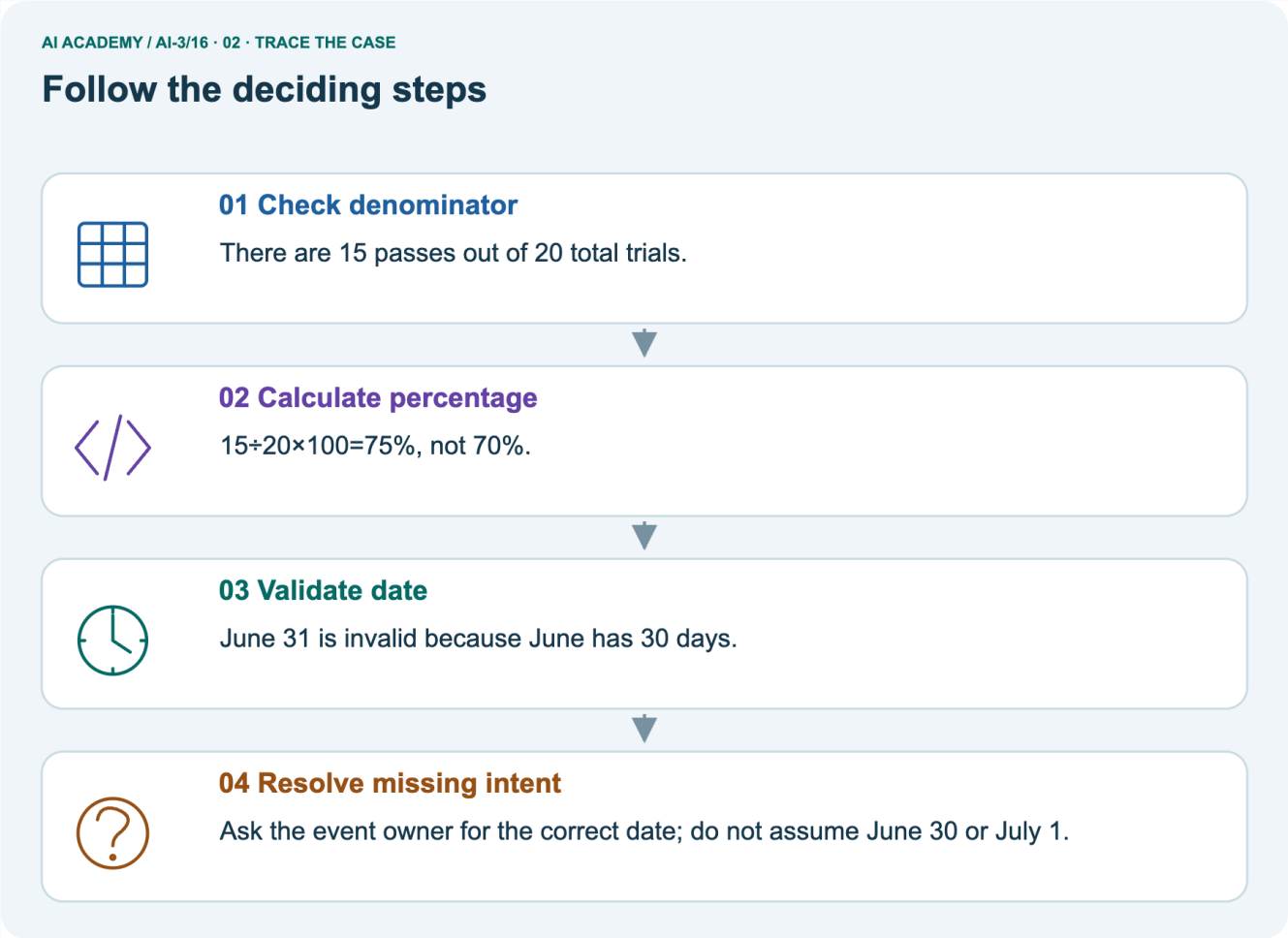
Draft: “**70%** passed; the meeting is on June 31.” June has 30 days. No authoritative record gives the intended meeting date.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

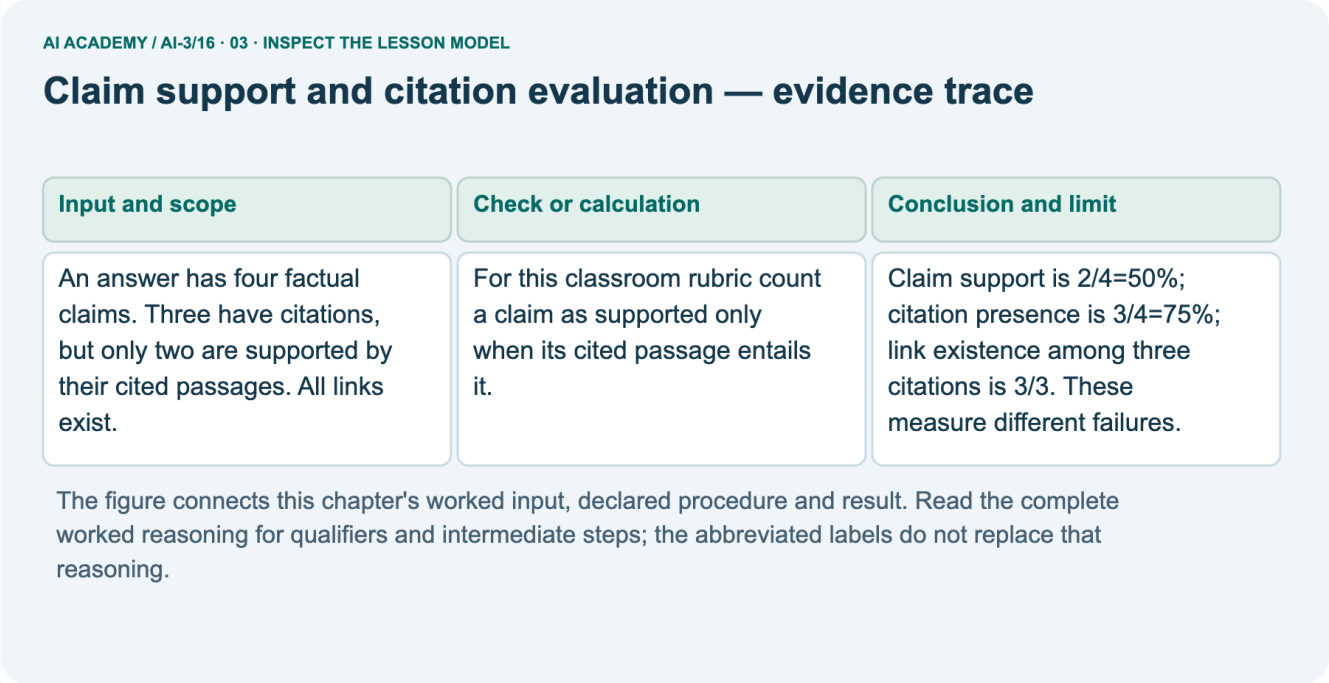
Check denominator: There are 15 passes out of 20 total trials. → Calculate percentage: 15÷20×100=75%, not **70%**. → Validate date: June 31 is invalid because June has 30 days. → Resolve missing intent: Ask the event owner for the correct date; do not assume June 30 or July 1.

> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Claim support and citation evaluation — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Claim support and citation evaluation — **evidence** trace. An answer has four factual claims. Three have citations, but only two are supported by their cited passages.

All links exist. | For this classroom rubric count a claim as supported only when its cited passage entails it. | Claim support is 2/4=50%; citation presence is 3/4=75%; link existence among three citations is 3/3.

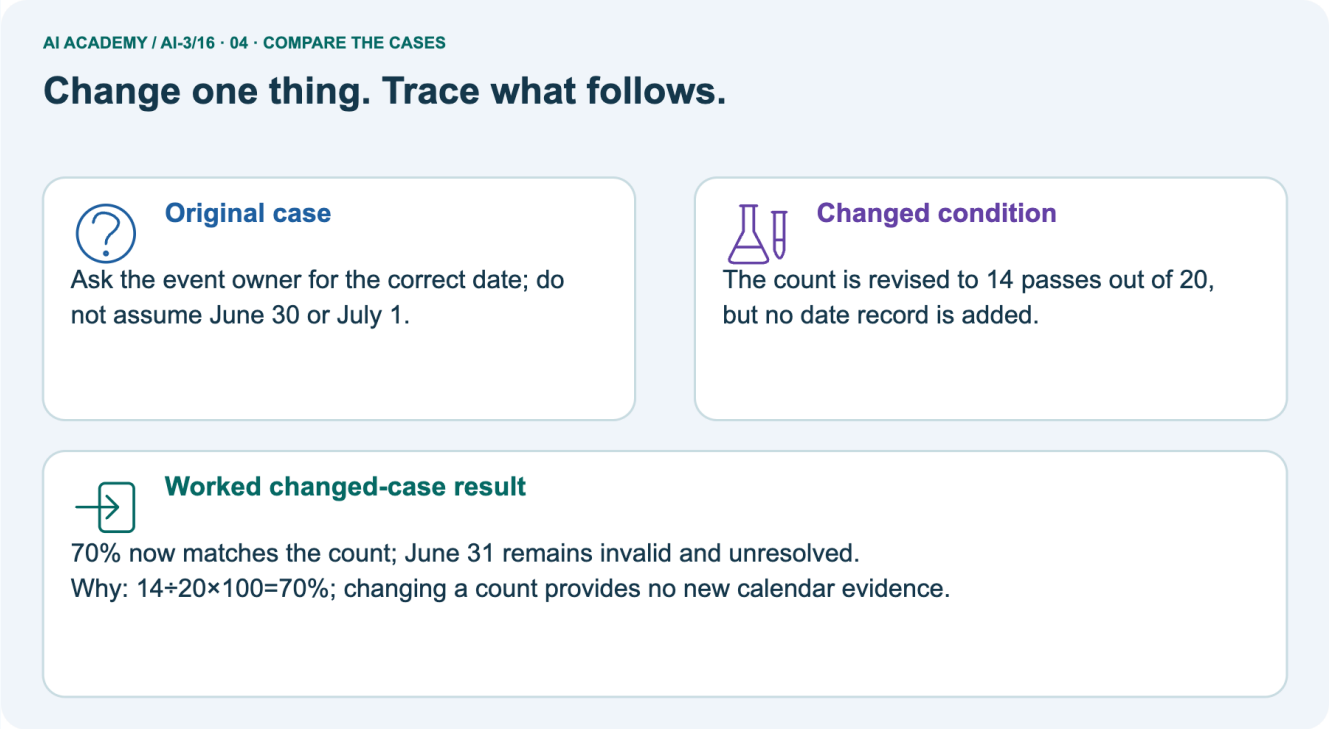
These measure different failures.

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Ask the event owner for the correct date; do not assume June 30 or July 1. Changed condition: The count is revised to 14 passes out of 20, but no date record is added.

Expected result: **70%** now matches the count; June 31 remains invalid and unresolved. Reason: 14÷20×100=**70%**; changing a count provides no new calendar **evidence**.

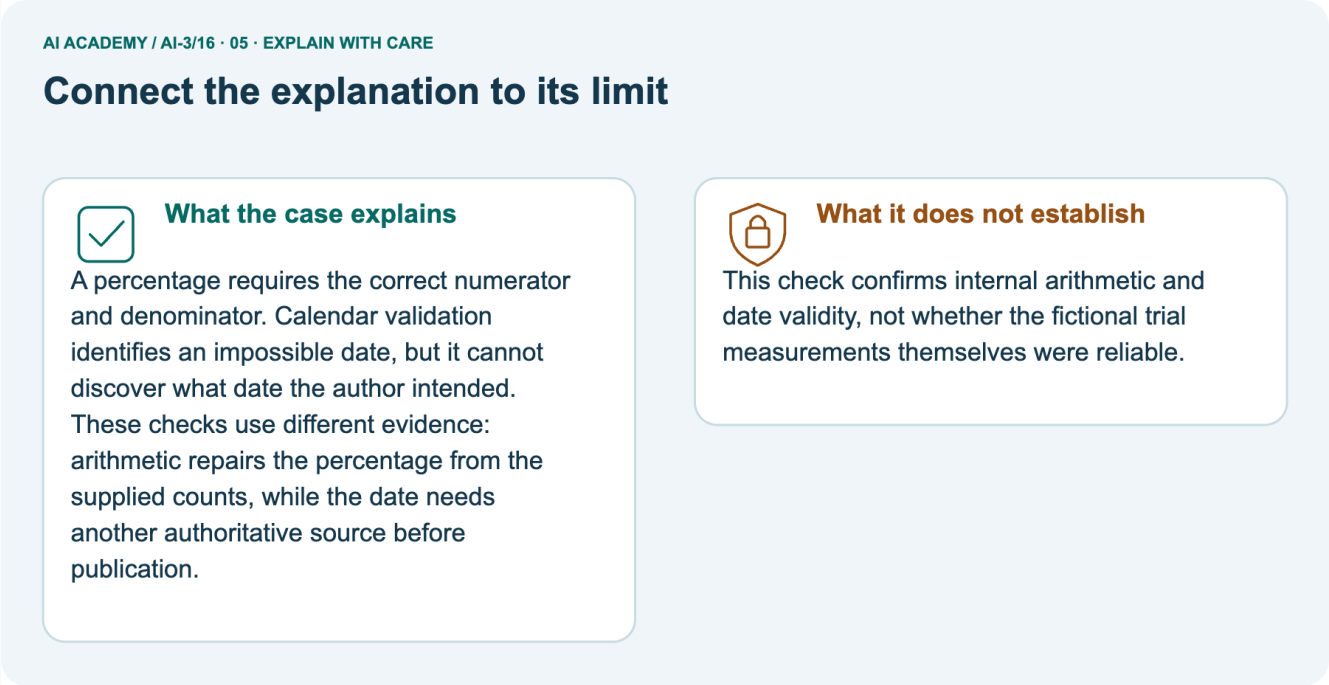
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: 14÷20×100=**70%**; changing a count provides no new calendar **evidence**.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A percentage requires the correct numerator and denominator. Calendar validation identifies an impossible date, but it cannot discover what date the author intended.

These checks use different **evidence**: arithmetic repairs the percentage from the supplied counts, while the date needs another authoritative source before publication.

Limit: This check confirms internal arithmetic and date validity, not whether the fictional trial measurements themselves were reliable.

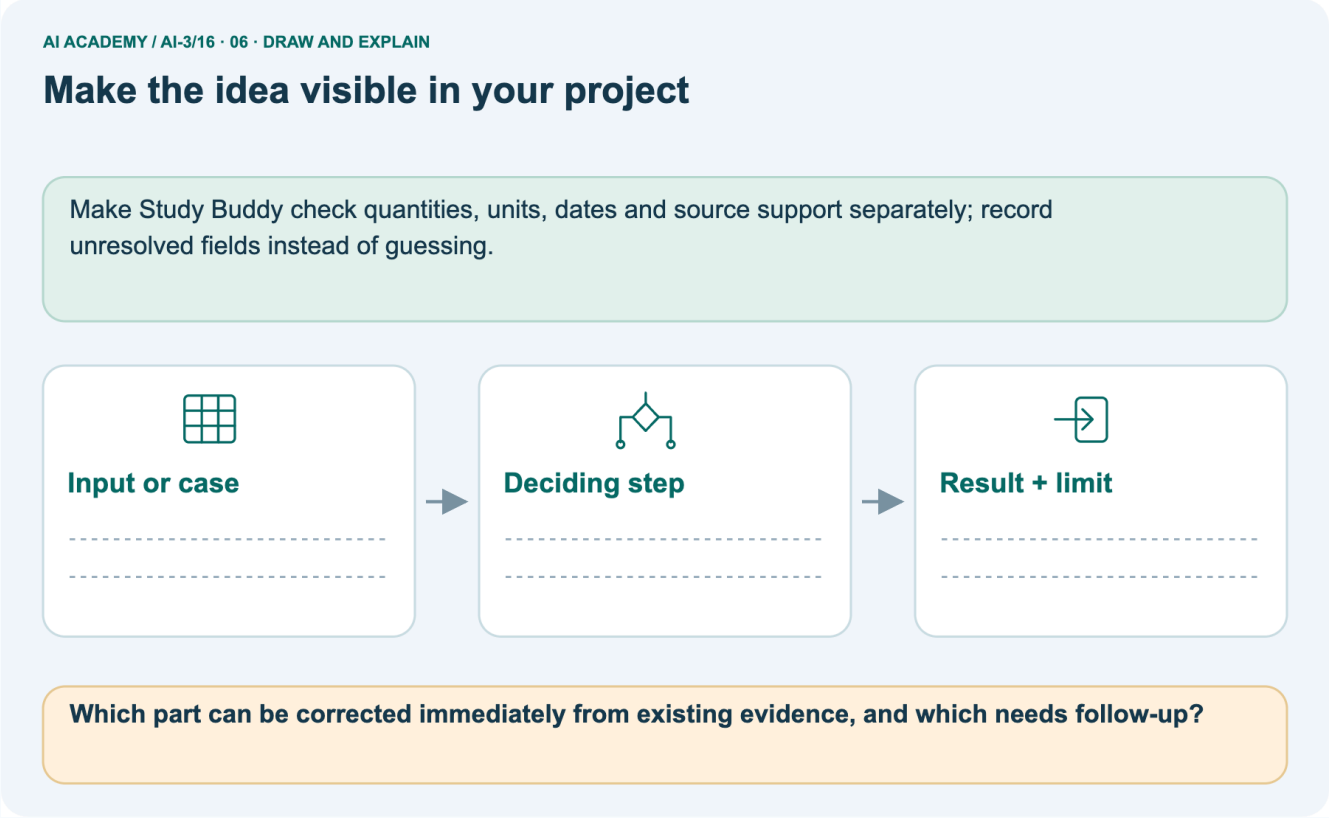
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: This check confirms internal arithmetic and date validity, not whether the fictional trial measurements themselves were reliable.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Make Study Buddy check quantities, **unit**s, dates and source support separately; record unresolved fields instead of guessing. Draw your **input**, deciding step, result and limit.

Supported question: Which part can be corrected immediately from existing **evidence**, and which needs follow-up?

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Correct percentage to 75% in the original case. The intended date needs confirmation from the responsible event source.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.